KALSHI VS. FED FUNDS FUTURES

Prediction Markets as Macro Data

Andres Blanco Prada
Panos Koutsoukos
Connor Rippert
Vish

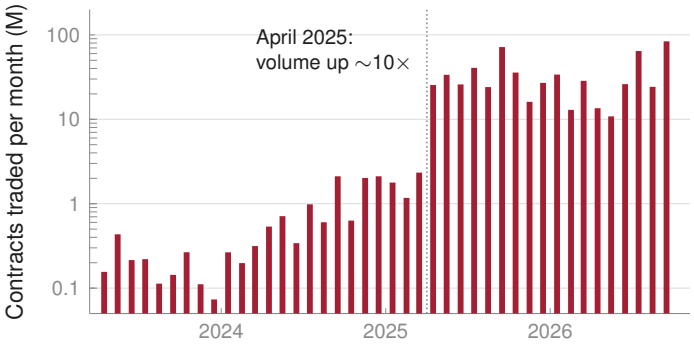
Progress update | October 2026

MIT Sloan School of Management

15.S06 AI and ML Research in Finance

Kalshi has become a deep market for Fed decisions

Our question: are these prices good forecasts, and are they better than fed funds futures?



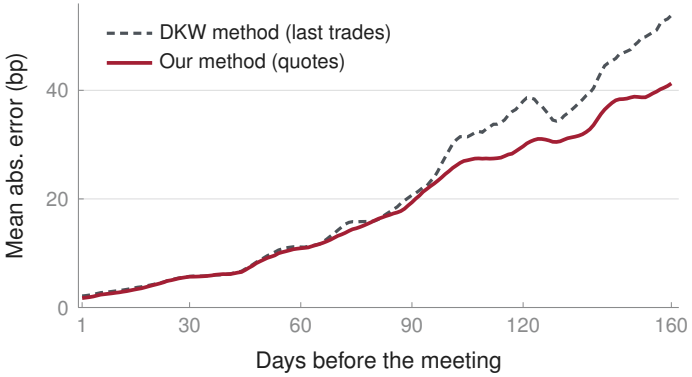
620M
contracts traded on FOMC decisions since 2023

28
settled FOMC meetings in the sample

1¢
median bid–ask spread

We replicate Diercks–Katz–Wright and improve on it

Kalshi-implied policy rate vs. the rate actually set, 41 FOMC meetings 2021–2026



41/41

meetings called correctly the day before

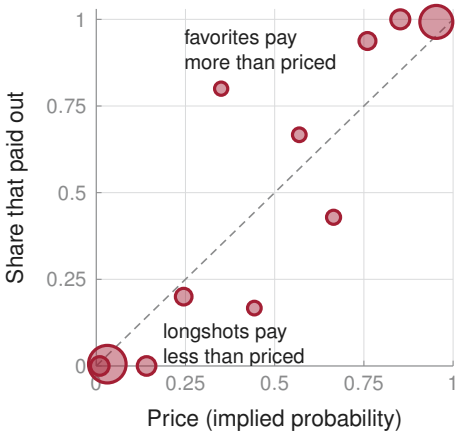
−14%

average error over 1–160 days (17.9 vs. 20.8bp)

Why: 40–45% of last trades are over a week old.

Fed-decision prices show a favorite–longshot pattern

Share of contracts that paid out, by price, 1–21 days before the meeting



0.5%

payout rate of contracts priced 1–10¢

(avg. price 3.1¢)

99.3%

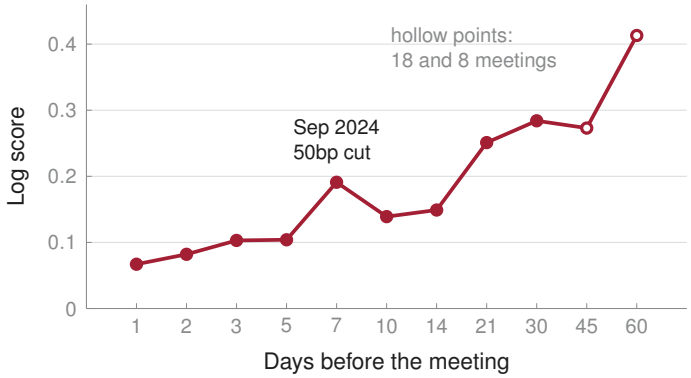
payout rate of contracts priced 90–99¢

(avg. 95.3¢)

Bubble size = forecasts. 632 forecasts, 28 meetings. Middle bins hold 2–6 meetings each. Formal test still to run.

Accuracy fades with horizon, and single surprises dominate

Log score of the full 5-outcome distribution, mean over meetings (lower is better)



0.07 → 0.28

log score, 1 day vs. 30 days out

0.05

median at 1–21 days, vs. mean 0.14

Uniform guess over 5 outcomes: 1.61.

Futures give a benchmark probability for every meeting

The WIRP-style method: from futures prices to outcome probabilities

Futures price
FF1–FF13, daily

→

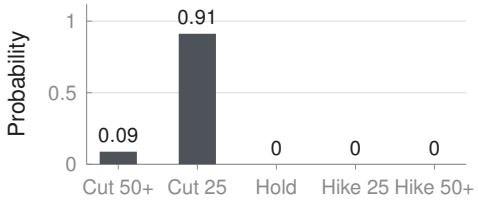
Avg. rate per month
$100 - P$

→

Rate after meeting
un-blend the meeting month

→

Expected move x
in 25bp steps



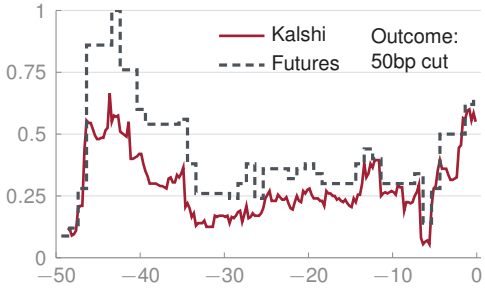
Example: 30 Jul 2024, for the 18 Sep meeting.
$x = -1.09$ steps $\Rightarrow$ 91% cut 25bp, 9% cut 50bp.

All mass sits on the two outcomes next to x, so the futures can never price the tails.

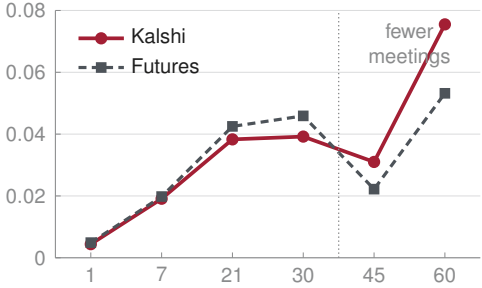
Kalshi tracks futures and is slightly more accurate to 30 days

Both read at 15:00 ET each day and scored against the actual decision

One meeting: Sep 2024, P(50bp cut)



All meetings: Brier score (lower is better)



Kalshi also puts more weight on moves beyond 25bp: +0.4pp the day before, +1.5pp at 31–60 days.

Appendix

Research questions and next steps

Questions

- **RQ1 Calibration.** Are Kalshi prices rational forecasts? Favorite–longshot bias? Better than futures and surveys? *In progress.*
- **RQ2 Price discovery.** How fast do prices absorb CPI, payrolls and FOMC news vs. futures? *Data ready.*
- **RQ3 Forecasting.** Use Kalshi distributions to predict CPI, unemployment and GDP. *Not started.*

Next steps

- Favorite–longshot regression, errors clustered by meeting.
- Formal Kalshi vs. futures test on per-meeting scores (e.g. Diebold–Mariano), split at April 2025.
- Rerun the futures comparison on refreshed candles (meetings after Sep 2025).
- RQ2 event study on 1-minute data around 8:30 and 14:00 ET.
- Add SPF and Bloomberg consensus benchmarks for CPI and payrolls.

Dataset: every Kalshi trade and 1-minute quote

Series	Events	Contracts	Trades	Volume	Since
Fed decision (KXFEDDECISION)	40	200	770,571	620M	Apr 2023
CPI m/m (KXCPI)	67	552	204,598	25.5M	Jul 2021
Core CPI m/m (KXCPICORE)	56	500	33,259	7.7M	Jul 2022
CPI y/y (KXCPIYOY)	49	713	241,413	37.9M	Dec 2022
Payrolls (KXPAYROLLS)	45	422	116,507	19.6M	Mar 2023
Fed funds ladders (KXFED)	52	660	$\sim$222,000	59M	Jul 2021

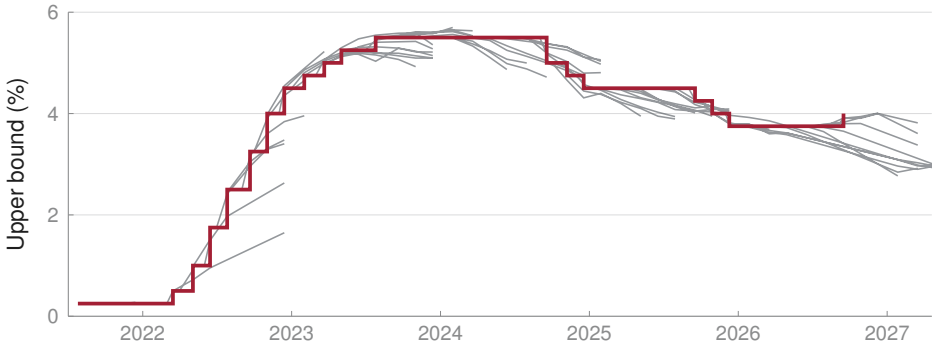
- 10.4M one-minute candles. 100% of 2,155 settled contracts match Kalshi's reported volume within $\pm$1%.
- Benchmark: Bloomberg FF1–FF13, EFFR and target range, daily 2000–2026.
- Fed outcomes: 19 holds, 5 cuts of 25bp, 1 cut of 50bp (Sep 2024), 3 hikes of 25bp.

Literature: what is known and what we add

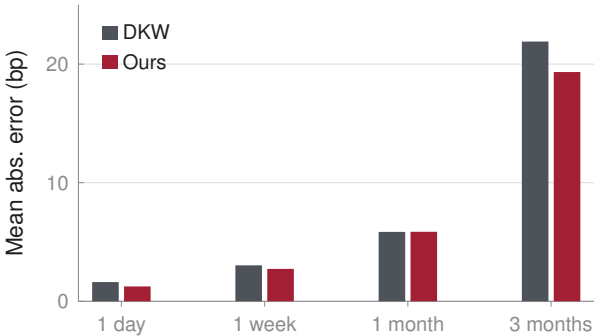
Paper	Finding	Gap we target
Diercks, Katz & Wright (2026)	Kalshi distributions for fed funds, CPI, GDP; CPI beats Bloomberg consensus	No Brier/log scores, no bias test, daily only
Bürgi, Deng & Whelan (2026)	Strong favorite–longshot bias overall; weak in economics contracts	Test on macro with clean specification
Bartlett & O'Hara (2026)	Microstructure of 41.6M Kalshi trades	No forecast evaluation
Wolfers & Zitzewitz (2004)	Contract design reveals mean, median, distribution	Early evidence: 16 obs.
Kelly et al. (2026)	Transformer stochastic discount factor	Only regularized ideas fit our sample (RQ3)

Kalshi-implied rate paths vs. the realized target

Each line: the path the market implied on one date; red steps: the actual upper bound



Replication accuracy by horizon



vs. NY Fed dealer survey

132 forecasts, 37 meetings: survey 19.1bp, Kalshi mode 17.0bp.

Method

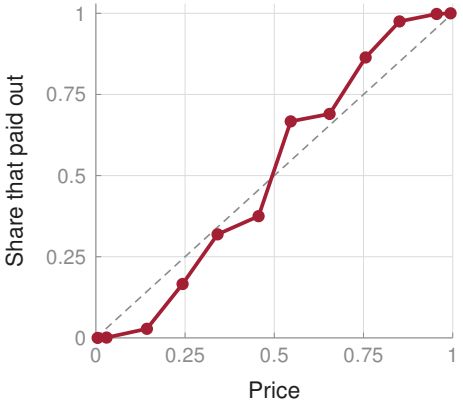
Mid quote if spread $\leq$ 20¢, else last trade (20–40¢) or dropped; weighted isotonic fit across strikes.

On DKW's 2022–25 window

19.9 vs. 22.7bp (1–160 days).

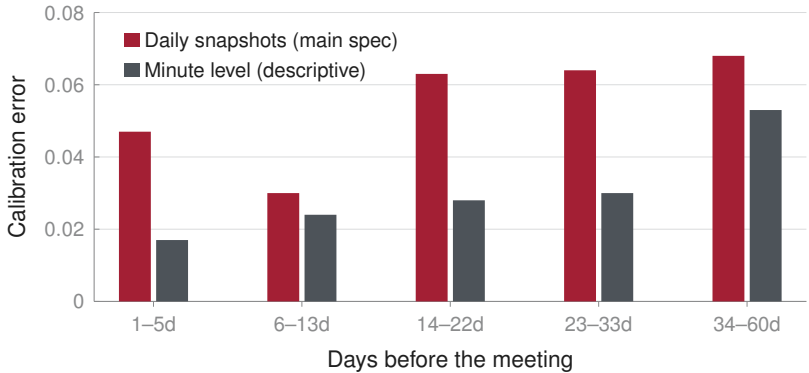
Minute-level reliability shows the same S-shape

Descriptive only: raw mid quotes, 0–60 days, includes meeting day



Calibration error grows with time to the meeting

Weighted calibration error, 20¢ bins; buckets hold equal shares of trades



Scores by horizon group, and the meetings that drive them

Horizon	Log mean	Log median	Brier mean
1–21 days	0.136	0.051	0.061
30 days	0.284	0.172	0.144
45–60 days	0.316	0.224	0.152

Meeting	Horizon	Log loss
Sep 2024 (50bp cut)	7 days	1.74
Sep 2024	30 days	1.13
Sep 2026 (25bp hike)	21 days	1.11
Sep 2024	21 days	0.88
Dec 2025	21 days	0.78

Per-contract binary log loss on the right. Days within a meeting are highly correlated, so the effective sample is about 28 meetings.

Specification fixed before any significance test

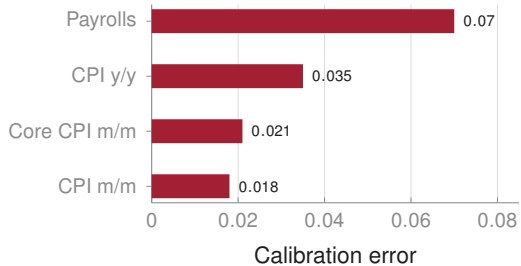
- Forecast: 16:00 ET bid–ask mid, $h \in \{1, 2, 3, 5, 7, 10, 14, 21, 30, 45, 60\}$ days before.
- Horizon groups: 1–21 days (main), 30, 45–60.
- Fed buckets normalized to sum to 1 (overround $\approx 1.5\%$); raw as robustness.
- Tick-floor quotes (below 1¢, 0¢ bid) excluded; included as robustness.
- Stale quotes kept and flagged; drop-stale as robustness. No spread filter.
- Scores per contract (log loss, favorite–longshot test) and per meeting (log score, Brier).
- Minute-level results are descriptive only.
- Voided contracts and one untraded duplicate Fed event dropped.

With about 28 meetings the test has low power, so the specification is fixed in advance.

CPI and payroll ladders are close to calibrated

All four series, 1–21 days before the release; 18,049 forecasts, 203 releases





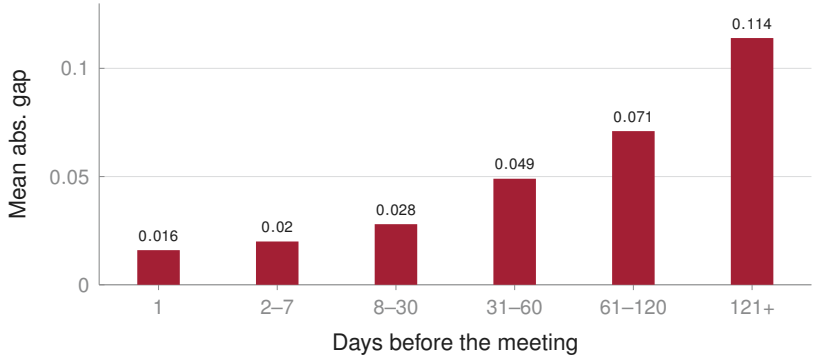
Payrolls is hardest: 1-day log loss 0.52 vs. 0.20–0.25 for CPI. Ladders are not monotone on 18–63% of release-days.

How futures-implied probabilities are computed

- FFn settles on the average EFFR of month $n-1$, so the implied average for month j is $\bar{r}_j = 100 - P_{j+1}$.
- A meeting on day d of an N-day month splits it: $\bar{r}_j = \frac{d}{N} r_{\text{pre}} + \frac{N-d}{N} r_{\text{post}}$.
  - Method A solves for r_{post} and amplifies noise by $N/(N-d)$.
  - Method B (preferred) takes r_{post} from the next meeting-free month.
- Meetings are chained; the start rate is the range midpoint plus a 10-day median EFFR spread.
- Expected move x in 25bp steps: with $n = \lfloor x \rfloor$, $p = x - n$, the move is n steps w.p. $1-p$ and $n+1$ steps w.p. p.
- Checks: all 2,909 (meeting, date) pairs sum to 1; Methods A and B differ by a median of 1.4bp.

Kalshi and futures agree closely near the meeting

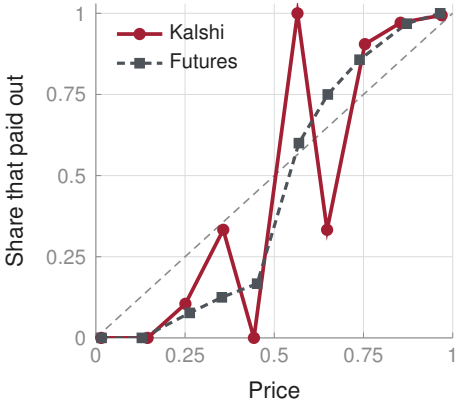
Mean absolute gap in probability, hold and ±25bp outcomes; 14,513 contract-days



Correlation falls from 0.998 (1 day) to 0.928 (121+ days).

Reliability: futures vs. Kalshi

Hold and ±25bp outcomes, 1–30 days pooled; bins from 0.3 to 0.6 hold 2–5 meetings



Scores: Kalshi vs. futures by horizon

Days before	1	7	21	30	45	60
Brier, futures	0.0049	0.0198	0.0425	0.0459	**0.0222**	**0.0532**
Brier, Kalshi	**0.0044**	**0.0191**	**0.0383**	**0.0392**	0.0310	0.0755
Log loss, futures	0.035	**0.081**	0.148	0.170	**0.113**	**0.192**
Log loss, Kalshi	**0.030**	0.084	**0.138**	**0.153**	0.142	0.253

Bold: lower (better) score. 28 meetings at 1–21 days, 8 at 60 days. Probabilities clipped to [0.005, 0.995].